

1.1 Modal Analysis User Guide: Frequency Domain Decomposition (FDD)

Frequency Domain Decomposition (FDD) and Enhanced FDD (EFDD) are among the most widely used tools for output-only modal identification. One variant, pre-scaled multi-dataset FDD, allows merging survey data from multiple datasets using reference sensors [1]. These methods are implemented in the Sensequake Cloud platform and can be accessed from the Analysis menu within a project.

The first step is to upload sensor data via the Project Data menu. The method supports multi-setup surveys, where one or more sensors act as references. Users can upload Sensequake Larzé data files in binary format or CSV files with sensor data organized in columns.

The FDD/EFDD process consists of two sequential steps: preprocessing and Frequency Domain Decomposition (FDD) to identify initial modal parameters, followed by Enhanced FDD (EFDD) to refine them with damping and improved mode shapes.

1.1.1 Preprocessing and FDD

The preprocessing step allows users to select the data source, apply basic signal processing, configure preliminary parameters for FDD, and perform the initial Frequency Domain Decomposition.

Before starting, note that the **FDD/EFDD Pre-Merging** method operates at the **group level only**, unlike the SSI method, which allows processing individual setups. Users should first verify the group and setup assignments in the **Project Settings** section.

The following steps will use the preprocessed data for the **FDD/EFDD analysis**. There are multiple sections of parameters in this step:

1.1.1.1 Main Settings

There are three parameters in the Main Settings section. The first, **FDD Frequency Resolution**, defines the spacing between spectral lines in the cross-power spectral density (CPSD). The maximum resolution is limited by the length of the recorded data. The default value is 0.1 Hz. A smaller resolution allows finer detection of frequency peaks but can make the results more sensitive to noise and increase computation time.

The second parameter is **Automatic Peak Picking Prominence**. This value sets the minimum prominence required for a peak to be detected in the SVD plot during the automatic peak picking process. The function uses the `scipy.signal.find_peaks` method, where *prominence* reflects how much a peak stands out relative to its surrounding data. A higher value will identify only the most distinct peaks, while a lower value may detect smaller peaks but risks including noise. For more details on automatic peak picking and the definition of prominence, users can refer to [find_peaks — SciPy v1.16.0 Manual](#).

The third parameter is the **Number of Singular Values to Plot**. This setting determines how many singular values are displayed in the EFDD and results sections. It only affects the visualization and does not influence the underlying mathematical analysis. Users should ensure that the number of singular values plotted does not exceed the product of the number of reference sensors and the number of data channels.

1.1.1.2 Data Channels

The **Data Channels** section allows users to select which sensor channels to include in the FDD/EFDD analysis. Two parameters can be adjusted: **Channels** and **Removed Sensor IDs**.

Users can choose from six channel options: **Vx**, **Vy**, **Vz**, **Ax**, **Ay**, and **Az**, where “V” represents velocity and “A” represents acceleration, and “X,” “Y,” and “Z” refer to the measurement axes. It is not recommended to mix velocity and acceleration data in the same analysis. If significant motion occurs mainly along one axis, selecting only that axis and muting the others may enhance the clarity of the mode shapes.

The **Removed Sensor IDs** setting allows users to exclude data from specific sensors—helpful when a sensor malfunctions or produces noisy data.

1.1.1.3 Signal Processing

The Signal Processing section allows users to modify the raw sensor data using three optional processing methods: **equalization**, **decimation**, and **cleanup**.

Equalization compensates for the natural decay of geophone signals below 4 Hz. This adjustment improves the quality of velocity signals in the low-frequency range, facilitating more accurate peak picking.

Decimation downsamples the signal after applying an anti-aliasing filter. This option speeds up calculations when higher-frequency signals are less relevant. For more information, users may refer to [scipy.signal.decimate — SciPy v1.11.3 Manual](#).

Cleanup removes signal portions that exceed the sensor’s operational limits. This helps mitigate unexpected interferences such as sudden impacts or sensor displacements during data collection. The cleanup function automatically filters out segments exhibiting abnormal acceleration or velocity levels across all sensors, maintaining synchronization throughout the dataset.

After setting the desired parameters, click the **Submit** button. The algorithm will merge PSD diagrams from multiple setups into a uniform PSD and calculate the corresponding SVD for the EFDD analysis.

1.1.2. EFDD

The EFDD analysis is performed after preprocessing and the initial FDD step, relying on the FDD results. EFDD consists of two main parts: parameter settings and peak picking.

Two key parameters must be adjusted: **Minimum MAC Threshold** and **Number of Singular Values**. These parameters are used to construct the bell function, which is central to EFDD analysis. The MAC (Modal Assurance Criterion) threshold controls how similar the mode shapes on the bell curve must be to the selected mode shape. The Number of Singular Values specifies how many of the first singular values (SVD lines) the algorithm uses to search for matching modes.

In the peak picking section, users identify peaks on the SVD diagram to run EFDD. Peaks can be selected either automatically or manually. Automatic peak picking uses the prominence parameter set in the preprocessing step. Alternatively, users can manually select peaks by clicking directly on the SVD diagram.

It is important to note that target frequencies should not be too low, as very low frequencies may cause EFDD to fail. Additionally, caution is advised when using automatic peak picking with acceleration data, since accelerometer signals often show prominent low-frequency peaks caused by sensor drift errors.

1. Amador, Sandro DR, and Rune Brincker. "Robust multi-dataset identification with frequency domain decomposition." *Journal of Sound and Vibration* 508 (2021): 116207.

1.2 Modal Analysis with FDD/EFDD: Getting Started

This page is intended to help users learn how to use the Dashboard through hands-on experience. A one-story building example and a three-story building example will be presented, and the modal analysis of each structure using FDD/EFDD will be demonstrated step by step.

Example: One-Story Building

We begin with a **one-story building** example. A 3D view of the structure is shown in Figure 1. The floor dimensions are 25 m by 31 m, and the story height is 8 m. Figure 1 also illustrates the global Cartesian coordinate system, node numbering, and overall geometry.

Note: the origin of the coordinate system is located at Node #1.

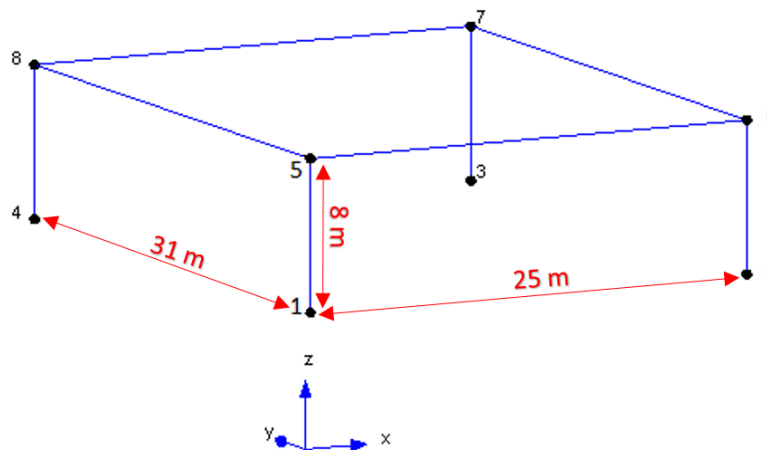


Figure 1: 3D View of the One-Story Building Model

Setting Up the Project

First, create a new project by clicking “+ NEW PROJECT” in the top-right corner of the homepage. From the dropdown menu, select “**Larzé Ambient Vibration System.**” This will take you to the project setup page.

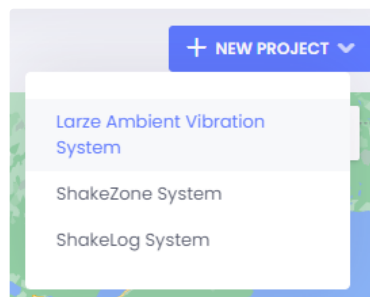


Figure 2: Creating a Larzé Project

The project setup page includes several sections: **Project Details**, **HAZUS Damage Estimation**, **Location**, **Time Zone**, **Units**, **Geometry**, and **Notes**. To begin, users can name the project in the **Project Details** section, then focus on the **Geometry** section for this example.

Within the **Geometry** section, multiple steps are required to define the structure's geometry. Information can be entered manually or uploaded via tab-delimited text files at each step.

The first step is to define the nodes. The **Nodes** section allows users to specify each node's ID, X, Y, Z coordinates, Sensor ID, and Setup ID. For the one-story building example, the **Nodes** section is shown in Figure 3.

Change Shape ×

ID	X	Y	Z	Sensor ID	Setup ID (0 for Reference Node)	+
1	0	0	0			 
2	25	0	0			 
3	25	31	0			 
4	0	31	0			 
5	0	0	8	5	1	 
6	25	0	8	6	1	 
7	25	31	8	7	1	 
8	0	31	8	8	1	 

Choose File No file chosen

Figure 3: Geometry — Nodes Settings

Next, define the lines. Each line connects two nodes. The line settings for the project are shown in Figure 4.

Change Shape ×

Node_1	Node_2	+
1	5	 
2	6	 
3	7	 
4	8	 
5	6	 
6	7	 
7	8	 
8	5	 

Choose File No file chosen

Figure 4: Geometry — Lines Settings

Next, define the surfaces. The current version of the Dashboard supports only triangular meshes; quadrilateral meshes are not supported at this time. The surface settings for this example are shown in Figure 5.

Change Shape ×

Node_1	Node_2	Node_3	+
5	6	7	 
5	7	8	 

Choose File

No file chosen

Figure 5: Geometry — Surfaces Settings

The remaining settings are related to HAZUS estimation and are not covered in this tutorial. Click the “✓” (**Update**) button to save your changes. Users can verify the geometry by clicking “**Preview**”, which displays the 3D model of the example (see Figure 6). Once satisfied, click the “**Submit**” button at the bottom of the page to send the changes to the server.

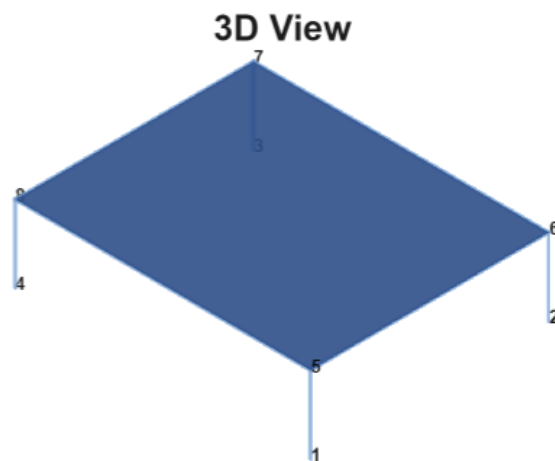


Figure 6: Geometry — 3D Structure Preview

The **Geometry** section also allows users to configure **Setup Info**, which refers to multi-setup tests. This part is skipped in the current example since only a single setup is involved. It will be discussed in more detail in the three-story building example.

Uploading the Data

Users can upload **.bin** files recorded from Larzé sensors. The Larzé Dashboard automatically detects the setup ID and sensor ID from the filename, so the filename should only be modified when necessary. The Dashboard also automatically reads the serial number, start time, duration, sampling frequency, and signal channels. The project data interface is shown in Figure 7. Note that the **Start Time** field may vary depending on the user’s time zone.

Project Data

Setup Number:

Setup 1

Upload

Upload Non-Sensequake Data

Download

Delete

☐ Sensor ID	Serial Number	Start Time	Duration (min)	Sampling Frequency (Hz)	Channels
☐ 5		Fri Dec 31 1999 19:00:00 GMT-0500 (Eastern Standard Time)	7	128	Vx
☐ 6		Fri Dec 31 1999 19:00:00 GMT-0500 (Eastern Standard Time)	7	128	Vx
☐ 7		Fri Dec 31 1999 19:00:00 GMT-0500 (Eastern Standard Time)	7	128	Vx
☐ 8		Fri Dec 31 1999 19:00:00 GMT-0500 (Eastern Standard Time)	7	128	Vx

Figure 7: Project Data for One-Story Building

Once data is uploaded, the **Setup Info** section in **Project Settings** → **Geometry** will update automatically. Users can assign a **Setup Group** number to each setup to facilitate processing. This step is skipped in the current example since only one setup exists.

FDD Modal Analysis

Once the project settings are complete, users can perform modal analysis by navigating to the **Analysis** panel on the left. The **Analysis** section offers three options: **Data Analysis**, **Damage Estimation**, and **Modal Analysis**.

For this example, select **FDD/EFDD** (not **FDD/EFDD Pre-Merging**) as the operational modal analysis (OMA) method. The **FDD/EFDD Pre-Merging** option is not used here because there is only one setup, so merging is unnecessary. The workflow for **FDD/EFDD** in single-setup cases is similar to the **FDD/EFDD Pre-Merging** process used in multi-setup cases, which will be demonstrated in the following example.

As outlined in the documentation, the FDD method consists of two main steps: **Preprocessing and FDD**, followed by **EFDD**. These steps will be detailed one by one in the next sections.

Preprocessing and FDD

First, select **Setup 1** from the Setup number menu. In the **Main settings**, the **FDD Frequency Resolution** is set to a default value of 0.1 Hz. Since the target modal frequencies generally lie between 0 and 20 Hz, this resolution offers a good balance for observing the overall frequency response in that range.

If users require more detailed PSD/SVD plots, the **FDD Frequency Resolution** can be refined to smaller values. However, finer resolution increases computation time and makes the PSD/SVD plots more susceptible to noise. There is also a lower limit on how fine the resolution can be: if set too fine, the Dashboard will display an error message in the “Response” console stating:

“Error while running preprocess step of setup 1: Error occurred during preprocessing: Frequency resolution too fine. Please adjust to a coarser df.”

The number of singular values is set to 3 by default. Note that this setting affects only the visualization and does not influence the underlying data or analysis.

For **Data Channels**, select **Vx**, since it is the only available channel in this example. No sensors are removed, and no reference sensors are specified. No additional signal processing is needed for this case.

Once all parameters are set, click the **Submit** button to proceed, as shown in **Figure 8**.

The screenshot shows a web interface for configuring preprocessing parameters. At the top, there is a 'Setup Number' dropdown menu set to 'Setup 1'. Below this, the 'Main Settings' section includes a 'Timelag (s)' input field with the value '0.5' and a 'Number of Singular Values to plot' input field with the value '1'. The 'Data Channels' section has a 'Channels' subsection with checkboxes for Vx (checked), Vy, Vz, Ax, Ay, and Az. Below this is a 'Removed Sensor IDs' text input field. Further down, there is a checkbox for 'Use reference sensor(s)'. The 'Signal Processing' section contains checkboxes for 'Filter', 'Use Equalization', 'Decimate', and 'Perform Cleanup', all of which are currently unchecked.

Figure 8: Preprocessing Parameters for One-Story Building Example

The screenshot shows a web interface for configuring modal analysis parameters. At the top, there is a 'Setup Number' dropdown menu set to 'Setup 1'. Below this, the 'Lowest Model Order' input field has the value '2' and the 'Highest Model Order' input field has the value '50'. The 'Clustering Threshold Factor' input field has the value '0.5'.

Figure 9: Modal Analysis Parameters for One-Story Building Example

In the FDD step, the Singular Value Decomposition (SVD) is computed. After SVD computation, users can proceed to the **EFDD** step to perform peak picking and run the EFDD analysis. The default **Minimum MAC Threshold** for bell curve identification is set to 0.9. The **Number of Singular Values** defines how many singular values the algorithm uses to construct the bell curve.

Additionally, in the **Advanced Settings: Log Fitting** section, there are two parameters: **Starting Peak** and **Number of Peaks**. These settings control the range of peaks that EFDD uses to estimate damping. For most cases, the default values are sufficient and can be left unchanged.

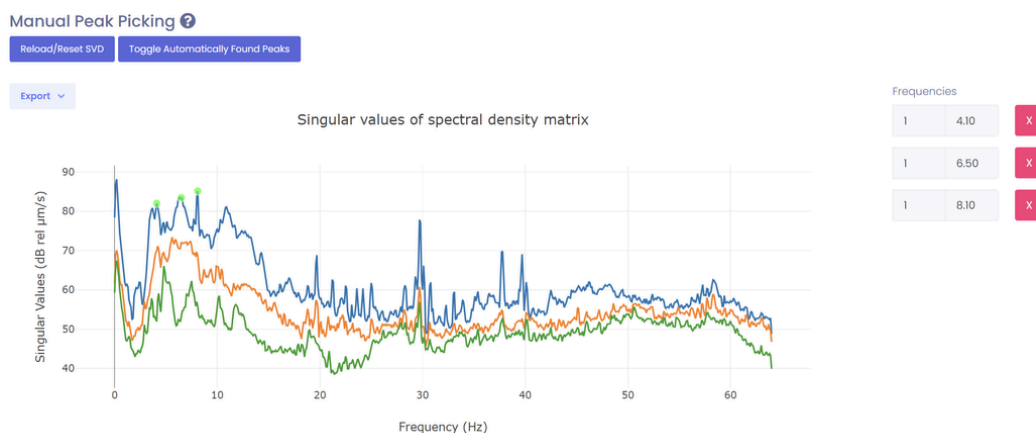


Figure 10

Users can also manually select peaks to analyze target modes by clicking directly on the peaks in the SVD plot. Alternatively, the **“Toggle Automatically Found Peaks”** button can be used to automatically detect peaks on the SVD plot, from which users can select their target modes. The frequencies of the selected peaks will be displayed on the right side of the interface. In this example, the first three peaks at 4.1 Hz, 6.5 Hz, and 8.1 Hz are chosen. Once the target peaks are selected, users can click **“Submit”** to proceed.

Results

Once all three steps of the FDD process are complete, the modal analysis result page will be displayed. Clicking **“Load Result”** on the Modal Analysis page will open the FDD results. The FDD results for the one-story building example, using the parameters described above, are shown in Figures 11 and 12.

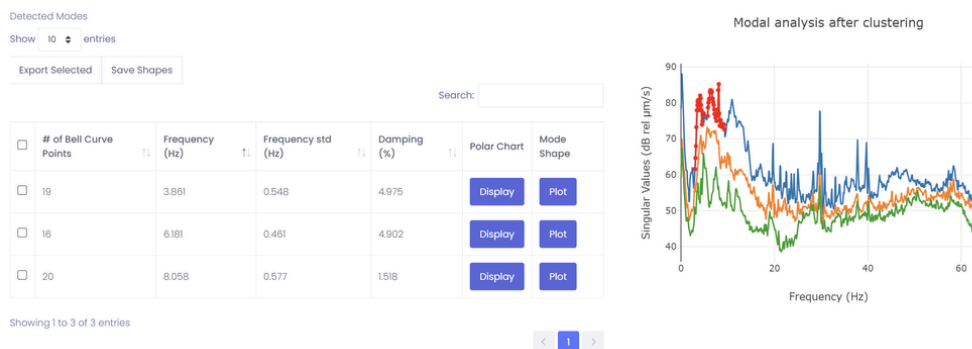


Figure 11

The table on the left displays key information for each mode cluster, including the number of bell curve points, the average and standard deviation of frequency, and damping. Two additional tools are available: a polar chart to visualize the mode phase—helpful for identifying whether a mode is physical or noise—and a mode shape animation.

In this demo, we examine the mode at 6.407 Hz. The cluster contains 23 poles, with an average frequency of 6.407 Hz (± 0.037 Hz) and average damping of 6.052% ($\pm 0.436\%$). As shown in Figure 11, the polar graph illustrates that the nodal values are tightly aligned near a phase angle of 0° , suggesting a clear, physical mode.

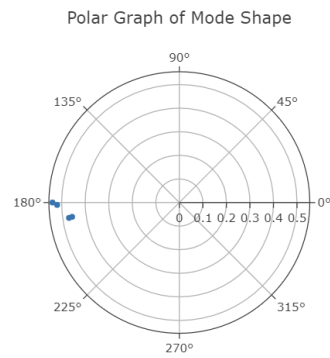


Figure 12

The mode shape is shown in Figure 13 and indicates an in-plane transverse mode in the x-direction. The top bar allows users to view the animation from five perspectives: four-view, front, top, side, and 3D. The animation can be paused and saved as a GIF.

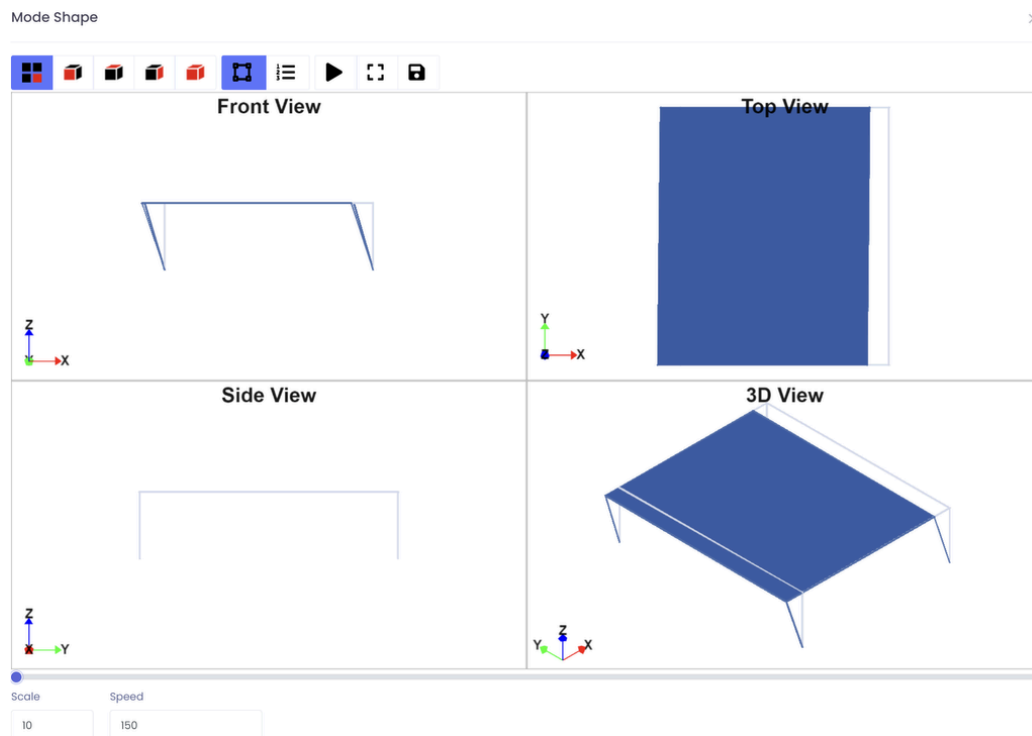


Figure 13

Example: Three-Story Building

The three-story building is the second example demonstrated in this document. This example uses a reference sensor and involves multiple tests. The demo will focus on the multi-setup test analysis. The geometry of the three-story building is shown in Figure 14.

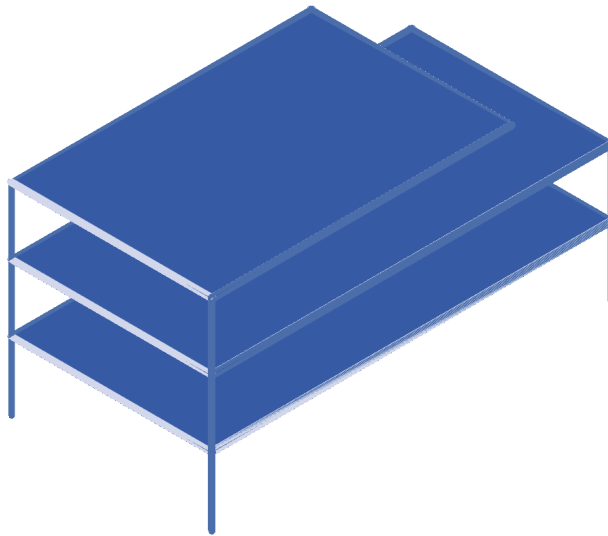


Figure 14: Geometry of Three-Story Building

Setting Up the Project

The steps to create the project are similar to those for the one-story building.

After creating the project, users can begin working in the **Geometry** tab for the three-story building. Defining the geometry involves a few additional steps, including specifying the setup and sensor IDs assigned to each sensor. In this example, the layout is predefined, and the mode shapes are known. In real-world applications, users should follow these guidelines for layout design:

- The reference sensor must remain the same across different setups.
- Only non-reference sensors may be moved between setups.
- Reference sensors should be placed at locations that are well excited by different modes.

Note: The steps for defining lines and surfaces are the same as those described for the one-story building example.

Uploading the Data

The **.bin** files should be uploaded to the Dashboard using the same method as before. The Dashboard will automatically organize the setups. As shown in Figure 15, the Dashboard recognizes two setups based on the filenames.

Once the data files are uploaded, the **Setup Info** section in **Project Settings** will update automatically. To perform modal analysis, ensure that the **Setup Group** is set to the same value for each setup, as Pre-merging FDD processes the entire group as a single unit.

Setup Number:

Select

Select

Setup 1

Setup 2

Figure 15:
Setup Numbers

Setup Info

Setup ID	Setup Group (0 for none)	Structure Name	Geometry Name	
1	1		INITIAL	
2	1		INITIAL	

Figure 16: Setup Info Section in Project Settings

Modal analysis

In this section, the EFDD analysis is performed. The **Preprocessing** and **FDD** steps are completed first, using the settings shown below. Note that only **Groups**—not individual setups—can be selected in the setup window.

The screenshot shows a software interface for 'FDD/EFDD Pre-Merging'. At the top, there are three tabs: 'SSI', 'FDD/EFDD', and 'FDD/EFDD Pre-Merging', with the last one being active. Below the tabs, there is a 'Step' dropdown menu set to 'Preprocessing + FDD'. Under 'Setup Number', a dropdown menu shows 'Group 1 (StructureName) [Setups 1, 2]'. The 'Main Settings' section includes 'FDD Frequency Resolution' set to 0.1, 'Automatic Peak Picking Prominence' set to 1, and 'Number of Singular Values to plot' set to 3. The 'Data Channels' section has checkboxes for Vx, Vy, Vz, Ax, Ay, and Az, with Ax, Ay, and Az checked. A 'Removed Sensor IDs' field is empty. The 'Signal Processing' section has three unchecked checkboxes: 'Use Equalization', 'Decimate', and 'Perform Cleanup'.

SSI FDD/EFDD FDD/EFDD Pre-Merging

Step

Preprocessing + FDD

Setup Number

Group 1 (StructureName) [Setups 1, 2]

Main Settings

FDD Frequency Resolution 0.1

Automatic Peak Picking Prominence 1

Number of Singular Values to plot 3

Data Channels

Channels

☐ Vx ☐ Vy ☐ Vz

☒ Ax ☒ Ay ☒ Az

Removed Sensor IDs

Signal Processing

☐ Use Equalization

☐ Decimate

☐ Perform Cleanup

Figure 17

Once the FDD step is complete, peak selection is performed manually in the EFDD step. The **MAC threshold** is set to **0.9**.

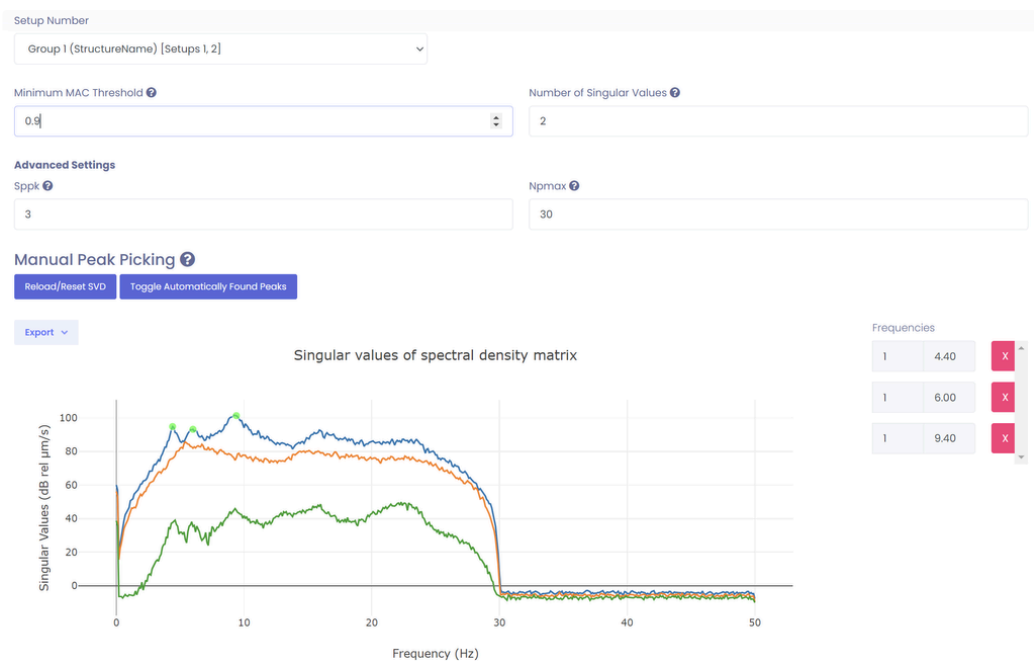


Figure 18

Results

The results section for multi-setup cases is as straightforward as for single-setup cases, since the algorithm has already merged the setups. The results page and the identified modes are shown below.



Figure 19

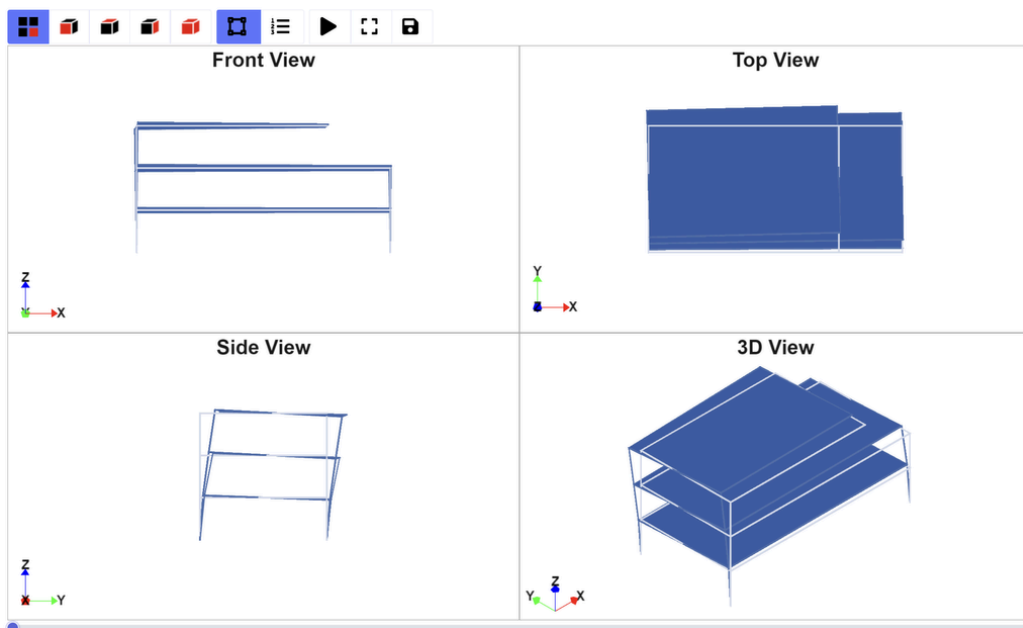


Figure 20

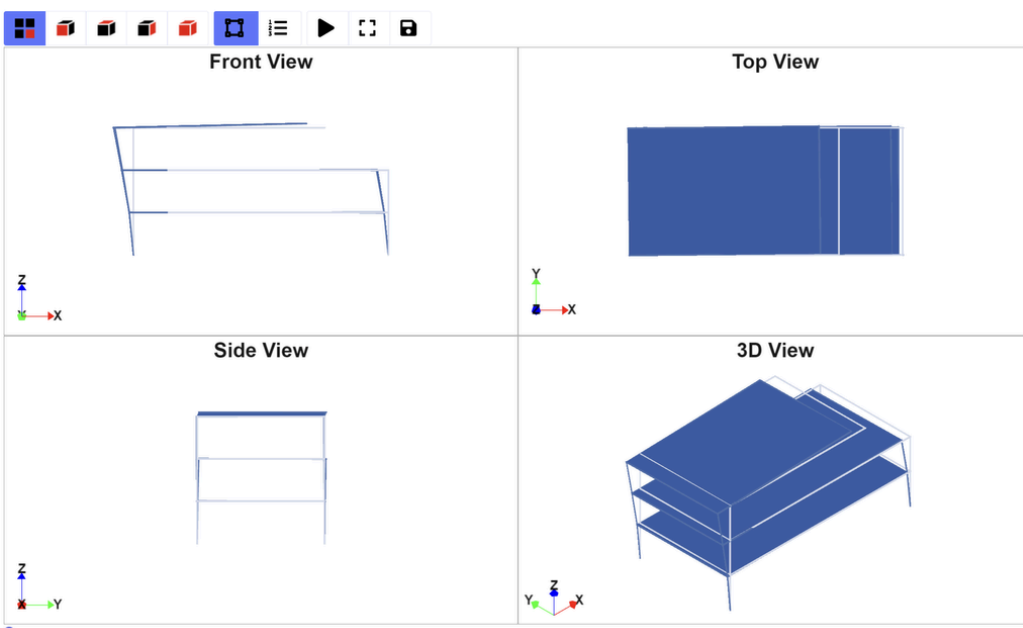


Figure 21

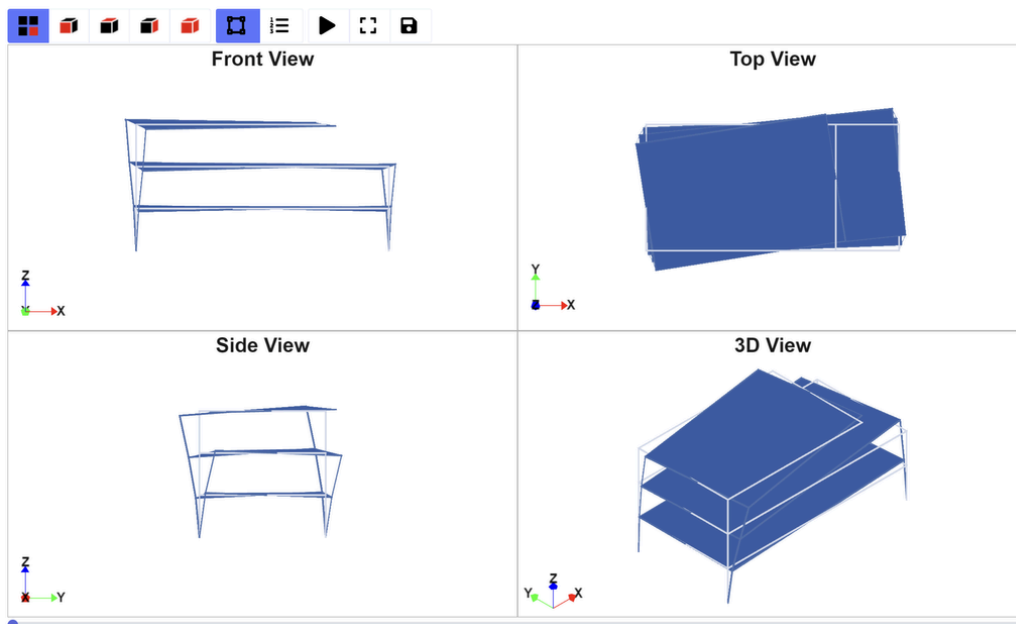


Figure 22